PROJECT DOCUMENT

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Team Members

T. Supriya

Santhanalakshmi N

Priyabharathi K

Kalaivani M

ServiceNow Project

1. Introduction

The Auto Ticket Classification using Flow Designer project is designed to automate the classification of IT support tickets in a school helpdesk environment. Students and teachers can submit support requests for common issues such as Wi-Fi connectivity, projector failures, password problems, and slow computers. The system analyses the ticket's short description and automatically assigns the appropriate category and subcategory using ServiceNow Flow Designer.

2. Problem Statement

The school IT helpdesk receives many incident requests from students and teachers. At present, IT staff manually review each request and select its category and subcategory. This process takes time, may lead to incorrect classification, and can delay ticket handling as the number of requests increases.

3. Objectives

- Automatically classify incidents when a ticket is created.
- Reduce the manual work performed by IT support staff.
- Improve ticket organization and routing efficiency.
- Maintain consistent category and subcategory values.
- Send an automatic email confirmation to the caller.
- Provide a no-code solution that is easy to maintain and extend.

4. Technologies Used

- ServiceNow – platform for creating and managing IT support tickets.
- Flow Designer – automates ticket classification and email notification.
- Update Sets – capture configuration changes for transfer between instances.

5. Requirement Analysis and Planning

The helpdesk requires a system that captures ticket details in a structured format and automatically identifies the type of issue from the short description. The solution must support dependent choices, assign both category and subcategory, notify the caller by email, and allow future maintenance.

5.1 Business Requirements

- Classify IT tickets based on the issue description.
- Assign Category and Subcategory automatically.
- Use dependent choice logic between Category and Subcategory.
- Send an email notification to the caller after ticket creation.
- Store ticket details in a structured table.
- Support future changes and enhancements.

6. Backend Development and Configuration

6.1 Create a New Update Set

1. Navigate to All > System Update Sets > Local Update Sets.
2. Select New and enter the name as Project Update Set.
3. Set State to In Progress and Application to Global.
4. Save the record.
5. Under Related Links, select Make This My Current Set.

6.2 Create the Custom Table

6. Navigate to All > System Definition > Tables.
7. Select New and set the table label to Incident Workflow.
8. Disable Create module if it is not required, and save the table.
9. Enable Auto Number in the table controls and save.
10. Open the table form and use Form Designer to add the required fields.

6.3 Field Creation and Data Types

Create the following fields in the Incident Workflow table. The Number field is configured as an auto-number field.

Field Label	Type	Values / Reference
Number	Auto Number	Generated automatically
Caller	Reference	User [sys_user]
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	Brief issue description
Description	String	Detailed issue description
State	Choice	New, In Progress, On Hold, Resolved, Closed
Assigned Group	Reference	User Group [sys_user_group]
Assigned To	Reference	User [sys_user]

Add the choice values to the Category, Subcategory, and State fields through the form or choice configuration, then save and reload the form.

6.4 Configure Category and Subcategory Dependency

The Subcategory field is configured as dependent on the Category field. This ensures that the available subcategory choices correspond to the selected category.

11. Open the Incident Workflow table form and configure the Subcategory dictionary entry.
12. Open Advanced view and enable Use dependent field.
13. Set Dependent on to Category and save.
14. Configure the dependent choice values as shown below.

Subcategory	Dependent Category
Wi-Fi	Network
Projector	Hardware
Forgot Password	Access
Slow Computer	Performance

7. Automation Using Flow Designer

Flow Designer is used to classify each new ticket without requiring the caller or IT agent to select the category manually. The flow is triggered when a record is created in the Incident Workflow table. It checks the Short Description for predefined keywords and updates the Category and Subcategory fields.

7.1 Create the Flow

15. Navigate to All > Process Automation > Flow Designer.
16. Select New and choose Flow.
17. Enter the flow name as Auto Classify School IT Tickets.
18. Set Application to Global and select Build Flow.

7.2 Configure the Trigger

19. Choose the Record Created trigger.
20. Select Incident Workflow as the table.
21. Set the condition to Category is empty.
22. Select Done.

7.3 Configure Classification Conditions

Add If and Else If branches. In each branch, use the Short Description data pill with the Contains operator. When a condition matches, add an Update Record action and set the relevant Category and Subcategory values.

Keyword(s) in Short Description	Category	Subcategory
Wi-Fi / WiFi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Forgot Password / Password / Login	Access	Forgot Password
Slow Computer / Slow / Hanging	Performance	Slow Computer

Use the keyword conditions consistently in the flow. If more than one keyword could match, place the more specific conditions first to avoid an incorrect classification.

7.4 Configure Email Notification

23. After the classification branches, add the Send Email action.
24. Set the recipient to the Caller's email address.
25. Use the subject: Your request has been submitted.
26. Add a message confirming that the support ticket has been created.
27. Save the flow and select Activate.

Sample email message:

> *Dear User,*
> *Your IT support request has been successfully created. The helpdesk team will review your request. Thank you.*

8. Testing and Validation

Testing is performed to confirm that the flow classifies tickets correctly, generates ticket numbers, stores field values, and sends email notifications to callers.

8.1 Test Scenarios

Test	Short Description	Expected Category	Expected Subcategory
1	WiFi not working in library	Network	Wi-Fi

2	Projector not turning on	Hardware	Projector
3	Forgot password for account	Access	Forgot Password
4	Computer is running slow	Performance	Slow Computer

8.2 Validation Checks

- Confirm that the required caller and short description details are entered.
- Check that the ticket number is generated automatically and is unique.
- Verify that Category and Subcategory are updated correctly after the flow runs.
- Confirm that reference fields point to the correct user or user group records.
- Check the email record under System Logs > Emails and preview the notification.
- Verify that only authorized users have access to view or update ticket records.

9. Deployment and Final Documentation

28. Navigate to All > System Update Sets > Local Update Sets.
29. Open Project Update Set.
30. Change its State from In Progress to Complete and save.
31. Under Related Links, select Export to XML.
32. The exported XML update set can be transferred to another ServiceNow instance and imported there.

10.Demo Link

https://drive.google.com/file/d/19dLHOU_zlKsGokFssjGMquCDDTHVEHb_/view?usp=drivesdk

11.Conclusion

The Auto Ticket Classification using Flow Designer project provides an automated way to classify school IT helpdesk tickets. By checking keywords in the Short Description, the system assigns the appropriate Category and Subcategory. Dependent choice fields help maintain consistent data, while the email action sends a confirmation to the caller after ticket creation. The use of Flow Designer provides a no-code approach that is easier to maintain and extend. The project can be enhanced in the future with automatic assignment to support groups, SLA tracking, and additional ticket categories.